`p8bprof_grpo_fast` — PRELIMINARY (steps 1–2 of 4)

dynamic token-budget micro-batching + chunked prefill, on partial rollout

Agentic-RL-CA rung-3 — `sync-partial-rollout-poc`

2026-07-30 05:5x CST; final 4-cycle update follows in this folder

TL;DR (preliminary)

The fast config is **healthy, OOM-free at the full 24,576 token budget**, and cuts update-side phases ∼**2.5**–**4**×; generation is unchanged (±7%, within the KV-fraction confound). Matched first-two-cycles wall-clock: **3,053 s vs 3,848 s** (1.26×; identical released counts 325/700). Projected 150-step GRPO arm: ∼**6.5 d** (from 8.9 d partial / 13.3 d vanilla). All speedup is attributable to **dynamic micro-batching**; chunked prefill contributed ≈0 at these prompt lengths. One **prominent flag** (§4): the `actor/entropy_loss` *metric* reads ∼2× lower under packing — a measurement artifact, not a policy change (evidence inside); all behavioral metrics match baseline within noise.

1. Launch config & deviations

Baseline `p8bprof_grpo_partial` + : `DYNBSZ=1` (`use_dynamic_bsz=True`, 24,576 tok/GPU, actor/old_log_prob/ref), chunked prefill ON with `max_num_batched_tokens=17408`. Same data/seed 0/256×5/4 cycles/partial args (`cycle_turns=8`, `max_age=4`); `ppo_mini_batch_size=512`; template, KV 0.5, remove-padding, balance-batch, grad-ckpt, sampling, algorithm settings untouched. **Deviations from request:** (a) `max_num_batched_tokens` 17,408, not 16,384 — the verl fork requires ≥`max_model_len`; (b) baseline ran at KV 0.6, fast at 0.5 (protocol default post-OOM-fix) — affects only the gen-side comparison (≲7%); (c) two aborted launch attempts before this one (config bugs, ∼50 min; no fallback used — OOM ladder never triggered).

2. Per-phase timing

	4-cycle means (final)			matched cycles 1–2	
	vanilla	partial	**fast (2-st. mean)**	partial	**fast**
gen (s)	3961	1591	1231	553 / 1753	587 / 1874
old_log_prob (s)	425	253	**47**	23 / 195	15 / 79
ref (s)	422	253	**48**	28 / 195	17 / 78
actor update (s)	2086	1262	**197**	125 / 967	51 / 342
total step (s)	7652	3728	1526	730 / 3118	672 / 2381
released traj	1280	932 avg	512 avg	325 / 700	325 / 700
step-s / traj	5.98	4.00	2.98	3.75	**2.98**
peak mem (GB)	85.2	85.1	85.3		no OOM @24576

3. Attribution

- **Dynamic micro-batching (DYNBSZ)**: actor update 2.8×, old_log_prob 2.5×, ref 2.5× at matched cycle 2 (967→342, 195→79, 195→78 s). This is the entire measured gain.
- **Chunked prefill**: gen 1753→1874 s at cycle 2 (+7%) — no benefit at our prompt lengths, and the sign is within the KV 0.6→0.5 confound. Keep (harmless) but attribute nothing to it.
- Generation now dominates: ∼80% of fast step time ⇒ **active-slot packing** (skip finished trajectories in gen) is the next real lever.

4. Sanity: training metrics vs baseline **one metric flagged**

Matched pre-update step 1: success 0.218 vs 0.237, response length 430 vs ∼500 tok, episode turns and partial counters in band; step-2 success 0.203 vs 0.184. All within temp-1.0 sampling noise. **Flag:** `actor/entropy_loss` reads 1.05/1.22 (steps 1/2) vs baseline 2.07/2.38 — ∼2× lower *at identical initial weights on step 1*, while the rollouts themselves are statistically indistinguishable. Since the entropy metric is computed during the old_log_prob recompute and aggregated over micro-batches, token-budget packing changes the *weighting of the average*, not the per-token log-probs; entropy has `coeff=0` and never enters the loss. Classification: **measurement artifact (high confidence)** — to be re-verified in the final report by recomputing token-weighted entropy from identical rollout dumps; if it were a true policy-distribution change it would have to appear in step-1 rollouts, and it does not.

5. Updated projections & recommendation (preliminary)

150-step arm	vanilla	partial	**fast (prelim)**
GRPO	13.3 d	8.9 d	∼**6.5 d**
turnPPO (est.)	20.3 d†	12.8 d	∼9–10 d (DYNBSZ factors applied to critic arm)

†pre-fix estimate; vanilla retry in flight. Two slots in parallel: the pair ≈10 d calendar at 150 steps, ≈5 d at 75 steps. **Recommendation:** adopt the fast config for the arms (gradient-identical, no observed behavioral drift, memory-safe); **active-slot packing is still worth building** — gen is now 80% of the step, and packing attacks exactly that without touching training data. Decide 150 vs 75 steps on scientific grounds, not throughput (6.5 d vs 3.3 d for GRPO).